

Poor Man's (College's) Proxmox Cluster

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What we needed

- Run three existing VMs, plus capacity for more
- Multiple nodes for continuity
- Failover
- Spares parts (for multiple nodes)
- NOT Broadcom
- Minimal cost

What we had

- Three VMs (one Windows, two Linux)
- One inherited node running VMWare
- No redundancy or fail over
- One complete set of spare parts

What we also had

- Around 100 HP EliteDesk 800 workstations recently removed from service
 - 4-core, 8-thread i7 @ 3.6GHz
 - 16GB RAM
 - 1TB SSD
 - 1GB/s ethernet
- Interest in Proxmox

Proxmox cluster requirements

- At least three nodes
 - We could come up with that
- Distributed ceph filesystem
 - We needed to add disks for that

What we did

- Got out three workstations
- Maxed out the RAM to 32GB
- Added a 2TB and 10TB spinning disk to each one
- Installed Proxmox VE for clustering on each one

How we configured it

- The 1TB SSD is the system disk
- The 2TB disk is local “scratch” storage
- The 10TB disk in each node is part of the ceph distributed file system

What we did to get ready

- Set up a stand alone Proxmox server that ran all three VMs
 - Allowed us to convert from VMWare to Proxmox, so importing into the cluster was trivial

What we got out of it

- It actually works!
 - Windows VM: 4 CPUs, 8GB RAM
 - Linux VMs: 4 CPUs, 4GB RAM
- Migration is simple and transparent
 - Allows for updates to nodes while VMs keep running
 - You can be ssh'ed in and not notice that the VM got moved
- We now have continuity in the event of a hardware failure
 - Using the High Availability Manager (ha-manager), failover happens quickly

What else we got out of it

- We have many spare parts
- Cost about \$340 per node
- Much experience with Proxmox
- Now we can ask for *moar mon-ay* for new nodes

What else are we planning

- Add one more node
- Create a new VM to replace our current (physical) Deep Freeze console



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What else do you want to know?

- Questions